

MTH212M ASSIGNMENT 2 HINTS WITH MARKING SCHEME

Note: Question 1 will be graded.

Question 1. (a) (1 + 1 + 1 marks) Prove or give a counter-example for each of the following statements.

- (i) Suppose that $i \rightarrow j$. If i is transient, then j is transient.
 - (ii) Suppose that $i \rightarrow j$. If i is aperiodic and positive recurrent, then j is aperiodic and positive recurrent.
 - (iii) Suppose that $i \rightarrow j$. If i is aperiodic and null recurrent, then j is aperiodic and null recurrent.
- (b) (1 + 1 + 1 + 1 + 3 marks) Let $M > 2$ be a positive integer. Let $\{X_n\}_{n=0}^\infty$ be a Markov Chain on the state space $S = \{1, 2, 3, \dots, M\}$ and with the T.P. matrix $P = (p_{ij})_{i,j \in S}$ such that $p_{ij} > 0, \forall 1 \leq i \leq j \leq M$ and 0 otherwise. Which of the following statements are necessarily true?
- (i) There is a single communication class.
 - (ii) There is an absorbing state.
 - (iii) All states have period 2.
 - (iv) P^T , the transpose of P , is stochastic.
 - (v) The sequence $\{X_n^2\}_{n=0}^\infty$ forms a Markov Chain.

Hints: (a)

- (i) Consider a Markov Chain with state space $\{0, 1\}$ and transition probability matrix $\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$.

Check that 0 is transient and 1 is recurrent. Here $0 \rightarrow 1$. (for similar counter-examples with details: 1 mark)

(ii) Since i is recurrent and $i \rightarrow j$, we have $i \leftrightarrow j$ (this argument: 0.5 mark). Consequently, using class property (this argument: 0.5 mark), j is aperiodic and positive recurrent.

(iii) Since i is recurrent and $i \rightarrow j$, we have $i \leftrightarrow j$ (this argument: 0.5 mark). Consequently, using class property (this argument: 0.5 mark), j is aperiodic and null recurrent.

(b)

(i) We have $p_{MM} > 0$ and $p_{Mj} = 0$ for $1 \leq j < M$. Hence, since the M -th rowsum is 1, we have $p_{MM} = 1$ and M is absorbing. Consequently, $\{M\}$ is a communication class and other

states will be in other communication classes (this argument: 1 mark). So there are more than one communication class.

- (ii) As discussed above M is absorbing. (details as above: 1 mark)
- (iii) The absorbing state has period 1. (this argument: 1 mark)
- (iv) We look at the M -th row sum of P^T . Here, (this argument: 1 mark)

$$\sum_{j=1}^M P_{Mj}^T = \sum_{j=1}^M P_{jM} = \sum_{j=1}^{M-1} P_{jM} + P_{MM} > 1.$$

Hence, P^T is not stochastic.

- (v) The sequence $\{X_n\}_{n=0}^\infty$ has the state space $S = \{1, 2, 3, \dots, M\}$. Consequently, $\{X_n^2\}_{n=0}^\infty$ takes values in the set $\{1^2, 2^2, 3^2, \dots, M^2\}$. (this statement: part marks 0.5)

Since $x \mapsto x^2$ is a bijection from $\{1, 2, 3, \dots, M\}$ to $\{1^2, 2^2, 3^2, \dots, M^2\}$, hence for all $x_0, x_1, \dots, x_{n+1} \in \{1, 2, 3, \dots, M\}$, (this argument: part marks 1)

$$\mathbb{P}(X_{n+1}^2 = x_{n+1}^2 \mid X_n^2 = x_n^2, \dots, X_0^2 = x_0^2) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0),$$

and

$$\mathbb{P}(X_{n+1}^2 = x_{n+1}^2 \mid X_n^2 = x_n^2) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

Using Markov property of $\{X_n\}_{n=0}^\infty$, we have (this argument: part marks 1)

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

Hence, for all $x_0, x_1, \dots, x_{n+1} \in \{1, 2, 3, \dots, M\}$,

$$\mathbb{P}(X_{n+1}^2 = x_{n+1}^2 \mid X_n^2 = x_n^2, \dots, X_0^2 = x_0^2) = \mathbb{P}(X_{n+1}^2 = x_{n+1}^2 \mid X_n^2 = x_n^2),$$

i.e. $\{X_n^2\}_{n=0}^\infty$ has the Markov property (this argument: part marks 0.5).

Question 2. ((2 + 2) + 6 marks) Let $\{X_n\}_{n=0}^\infty$ be a Markov Chain on the state space $S = \{0, 1, 2, 3, 4, 5\}$ and with the T.P. matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

- (a) Find all transient states and recurrent classes.

(b) For all transient states i and all recurrent classes C , compute

$$\mathbb{P}(X_n \in C \text{ for some } n \mid X_0 = i).$$

Hints: Using values of p_{ij}^n , verify that $\{0, 1\}$, $\{2, 3\}$, $\{4, 5\}$ are the communication classes. Further, $\{0, 1\}$, $\{2, 3\}$ are recurrent classes and 4, 5 are the transient states.

Take $\beta_{\cdot C} = \left(\frac{\mathbb{P}(X_1 \in C \mid X_0=4)}{\mathbb{P}(X_1 \in C \mid X_0=5)} \right)$ with $C = \{0, 1\}$ for example. Here, $\beta_{\cdot C} = \left(\frac{\frac{1}{4}+0}{\frac{1}{6}+\frac{1}{6}} \right) = \left(\frac{\frac{1}{4}}{\frac{1}{3}} \right)$.

As discussed in class, $Z = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{6} & \frac{1}{6} \end{pmatrix}$ and

$$\alpha_{\cdot C} = \begin{pmatrix} \mathbb{P}(X_n \in C \text{ for some } n \mid X_0 = 4) \\ \mathbb{P}(X_n \in C \text{ for some } n \mid X_0 = 5) \end{pmatrix} = (I - Z)^{-1} \beta_{\cdot C}$$

Complete the calculation for all choices of C and all choices of transient states.